

Tianyun Zhong

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Research Interests: Large Language Models · Long-Horizon Agents · Post-Training

Education

Institute of Software, Chinese Academy of Sciences (ISCAS)

Sep. 2024 – Present

M.S. in Electronic Information

- Advisors: Xianpei Han, Le Sun | GPA: **3.88/4.00** | Honor: Outstanding Student, UCAS

University of International Relations

Sep. 2020 – Jun. 2024

B.S. in Information Management and Information Systems

- Major Average: **91.43/100** | Honors: Beijing Outstanding Graduation Thesis (1 of 2 university-wide); Beijing Outstanding Student Leader

Publications

[SCI-AGENT-EXAM] **Benchmarking Agents on Scientific Research Tasks**

Tianyun Zhong, Wei Wang, Jialin Li, Jianhong Tu, Liyao Li, Chuqiao Kuang, Tianshi Zheng, Songchi Zhou, Weiqi Zhai, Qile Jin, Jiajia Li, Jiasheng Zheng, Boxi Cao, Yaojie Lu, Hongyu Lin, et al.

Under Review

- Developed a workflow-oriented benchmark for scientific agents with **120 executable tasks across 11 scientific domains**, covering **7 research task types** spanning scientific reasoning, coding, tool use, analysis, and interpretation.
- Designed a **hybrid evaluation framework** that jointly measures execution and presentation, and conducted trajectory-level analyses of agent efficiency and solution stability.

[LongGameBench] **Are LLMs Long-Termist? Evaluating Long-Horizon Optimization in Roguelike Environments**

Wenjuan Zhang*, Tianyun Zhong*, Wei Wang, Yaojie Lu, Li Wenbo, Yuzhen Shi, Boxi Cao, Shiwei Ye, Hongyu Lin, Bing Zhao, Hu Wei, Lin Qu, Le Sun, Xianpei Han.

Under Review

- Introduced **Long-Horizon Optimization** and developed a unified Agent-Environment benchmark based on Roguelike environments; evaluated **15 frontier LLMs** across **6 agent harnesses** using **1,000+ complete trajectories**.

[HypoArena] **Before the Action: Benchmarking LLMs on Prospective Hypothesis Discovery**

Tianyun Zhong, Wangyi Jiang, Wei Wang, Xuanang Chen, Yaojie Lu, Shiwei Ye, Yuzhen Shi, Boyu Yang, Jinghang Wang, Han Li, Haiyang Yu, Yongbin Li, Hongyu Lin, Le Sun, Xianpei Han.

Arxiv | [\[Code\]](#) [\[Data\]](#) [\[Paper\]](#)

- Proposed **Prospective Hypothesis Discovery** and the **Retrospective Context Regression** paradigm for reverse data synthesis; designed an Arena-style pairwise evaluation framework for open-ended hypothesis generation.
- Led the end-to-end data pipeline and fine-tuned a dedicated Judge model with Qwen models, improving evaluation accuracy by approximately **15%**.

[AOE] **Beyond Isolated Dots: Benchmarking Structured Table Construction as Deep Knowledge Extraction**

Tianyun Zhong, Guozhao Mo, Yanjiang Liu, Yihan Chen, Lingdi Kong, Xuanang Chen, Yaojie Lu, Hongyu Lin, Shiwei Ye, Shanshan Jiang, Bin Dong, Le Sun, Ben He, Xianpei Han.

AAACL-IJCNLP 2026 | [\[Paper\]](#)

- Built a structured knowledge extraction benchmark for long documents with an average length of **171K tokens**; designed both Offline LLM and Online Agentic settings to investigate long-context retrieval, reasoning, and structured knowledge construction.
- Developed benchmark construction standards and annotation protocols, and implemented an Online Agent pipeline based on RAG and dynamic Agent navigation.

XMC-Agent: **Dynamic Navigation over Scalable Hierarchical Index for Incremental Extreme Multi-label Classification**

Yanjiang Liu, Tianyun Zhong, Yaojie Lu, Hongyu Lin, Ben He, Shuheng Zhou, Huijia Zhu, Weiqiang Wang, Zhongyi Liu, Xianpei Han, Le Sun.

ACL 2024 Findings | [\[Paper\]](#)

- Developed a hierarchical-index-based dynamic navigation approach for scalable incremental extreme multi-label classification, incorporating reasoning, planning, and backtracking for efficient label retrieval.

Intern Experience

Alibaba Group – Tongyi

Research Intern

Oct. 2025 – Present

- **SWE-Science – Scientific Coding Agent Benchmark:** Built a multi-domain Agent benchmark covering **Bioinformatics, Scientific Computing, Chemistry, Data Science, and Physics**, with 240 tasks spanning Tool Orchestration, Code Generation, Hypothesis Validation, and Error Diagnosis. Developed an executable scientific-computing environment based on **Harbor**, integrating Python/R environments and domain-specific database APIs, together with Programmatic Evaluation and Agent-as-a-Judge for fine-grained trajectory assessment.
- **QoderCodingBench – Real-World Coding Agent Benchmark:** Addressed the limited coverage of feature development in SWE-Bench by constructing feature-level Coding Agent tasks, with **84%** feature tasks. Built an end-to-end pipeline from real user sessions and intent reconstruction to sandbox generation and Harbor task construction, and developed session-level batch evaluation with Agent-as-a-Judge to provide structured feedback for Agent optimization.
- **LongGameBench – Long-Horizon Agent Decision-Making** *Submitted to AAAI 2027 – Co-First Author*
Investigated long-horizon planning, resource management, and continual decision-making in Agent systems. Designed an **MCP-based Agent–Environment interface** for Roguelike environments and analyzed **1,000+ trajectories** from **15 frontier LLMs × 6 Agent harnesses**, identifying systematic failure modes in long-horizon optimization.
- **HypoArena – Prospective Hypothesis Discovery** *Submitted to NeurIPS 2026 – First Author*
Studied the ability of Generalized Agents to formulate high-quality hypotheses under sparse feedback. Led the pipeline from expert-report parsing and automated task generation to preference annotation, and fine-tuned a dedicated **Qwen3.5-based Judge** with pairwise preference evaluation, improving Judge accuracy by approximately **15%**.

Baidu – PaddleNLP

Research Intern

Large-Model-Based Vector Retrieval

- Adapted LLaMA and other decoder-only LLMs into embedding models through **contrastive learning**, building the complete training, retrieval, and evaluation pipeline from scratch.
- Proposed a **layer-contribution-based pruning strategy** for efficient embedding inference, achieving approximately **20% inference speedup** while maintaining retrieval performance. The core implementation was merged into the official PaddleNLP repository.  [\[Code\]](#)

Honors & Awards

- **Academic Service:** Reviewer for ACL Rolling Review (ARR), 2026
- Outstanding Student, University of Chinese Academy of Sciences, 2025
- Beijing Outstanding Graduation Thesis, 2024 *DocAgent: An AI Agent for Document Database Retrieval and QA* (1 of 2 university-wide)
- Beijing Outstanding Student Leader, 2024
- First-Class Scholarship, University of International Relations (multiple years)

Research & Technical Skills

- **Research Areas:** LLM Agent reasoning and planning, Long-Horizon Agents, long-context understanding, RAG, Agent–Environment Interaction, LLM-as-a-Judge evaluation.
- **Post-Training & Modeling:** LLM SFT, preference optimization (DPO), contrastive learning; familiar with PPO and GRPO.
- **Research Methods:** Research ideation, benchmark design, synthetic data construction, preference data annotation, experimental design, trajectory analysis, automated evaluation, and academic writing.
- **Engineering:** Python, PyTorch, Transformers, vLLM, FAISS, Elasticsearch, Docker, Git; experienced in building large-scale automated evaluation and data-processing pipelines.
- **English:** TEM-8; proficient in academic literature reading, technical communication, and academic writing in English.